

SALEXA — MOBILE APPLICATION

Complete documentation (Android + iOS)

PART 0 — READ THIS FIRST

0.1 What the mobile application is

The **phone app used by the person on the ground** — the field sales agent, delivery rider, service technician, loan officer. It is their whole working day: clock in, follow the route, meet customers, take orders, collect money, claim expenses, clock out.

It is not a small version of the web app. The web app is for watching and managing. The mobile app is for **doing**.

0.2 The three design rules that decide if this app succeeds

1. **It must work with no internet.** Agents work in villages, basements, and dead zones. Every action saves on the phone first and syncs later.
2. **It must not eat the battery.** This is the single biggest complaint against competing apps (FieldSense users report battery drain and phone heating). Our whole tracking design exists to prevent this.
3. **It must be brutally simple.** Two taps to punch in. Three to log a visit. An agent should never need training.

0.3 One app, two product levels

Product bought	What the agent sees
Agent Live Tracking	Punch in/out, my route, my attendance, diagnostics
Sales & Field Force	All of the above + beat plan, visits, orders, payments, expenses, forms, targets

Screens simply appear or hide based on what the company purchased.

PART 1 — EVERY SCREEN (in the order the agent meets them)

M1. Splash / auto-login

- Logo, checks saved login, goes straight to Home if already logged in.

M2. Login

- Company code (optional), phone or email, password. OTP option.
- "Forgot password". Big **Log in** button.

M3. Permission setup wizard (*the most important onboarding screen*)

- A guided, 5-step flow. **Never fires a raw system popup without explaining first.**
- Each step: plain-language reason → then the system prompt.
- Detects the phone brand (Xiaomi / Oppo / Vivo / Realme / Samsung) and shows **brand-specific instructions with a deep-link** to the right settings page.
- Full details in **Part 2**.

M4. Home / My Day (*the main screen*)

- **Big Punch In / Punch Out button** in the centre.
- Today at a glance: on-duty status, hours worked, visits done vs planned, orders booked.
- **Pending-sync badge** — how many items are waiting to upload.
- Quick buttons: Take order · Add expense · My route.

M5. Beat plan / My route (*Sales product*)

- Today's customers in visit order — map + list.
- Each stop: name, address, distance, status (pending / done / skipped).
- **Navigate** button → opens turn-by-turn directions.

M6. Customer detail

- Contact, address, phone (tap to call), GST, credit limit.
- Their past orders, outstanding amount, notes.

- Buttons: **Check in** · **Take order** · **Collect payment**.

M7. Check-in / check-out

- GPS-verified: confirms the agent is actually at the shop (inside the geofence).
- Optional photo. Visit timer starts.
- Blocks check-in if a **fake-GPS app** is detected.

M8. Visit report

- Outcome (met / not available / closed), person met, notes.
- Next action + follow-up date.
- Photos; attach a custom form.
- **Saves offline.**

M9. Take order

- Product search → quantity → price and tax auto-filled.
- Live running total. Stock warning if short.
- Discount within the agent's allowed limit.
- **Save (works offline)** → queued for sync.

M10. My orders

- Orders I booked and their status: pending approval → confirmed → dispatched → invoiced → paid.

M11. Collect payment

- Amount, mode (cash / UPI / cheque), against which invoice.
- Generates a receipt; **works offline.**

M12. Expenses

- New claim: type (travel / food / stay), amount, **receipt photo**, notes.
- **Auto-distance** pulled from the tracked route (for travel allowance).
- Location-validated. List of my claims + approval status.

M13. Attendance & leave

- My attendance history (calendar/list): in, out, hours.
- **Apply for leave** → type, dates, reason. See balance and approval status.

M14. Custom forms

- Any form the admin published: surveys, audits, competitor info, feedback.
- Photo, signature, dropdown, GPS fields. **Works offline.**

M15. My targets

- Target vs achieved this month: orders, value, visits.

M16. Messages / notifications

- Manager messages, task alerts, approval results, reminders.

M17. Sync centre (*critical for low-network areas*)

- Shows: items pending upload, what failed and why, last successful sync.
- **Sync now** button.
- Nothing is deleted from the phone until the server confirms receipt.

M18. Diagnostics / self-check

- One tap: checks GPS, permissions, background running, battery settings, storage, network.
- Green ticks / red crosses with a **Fix it** button that opens the right settings page.
- The agent can repair their own phone without calling support.

M19. Profile & settings

- My info, team, manager. Language. Notification preferences. App version. **Log out.**
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PART 2 — PERMISSIONS (what we ask for, and why)

2.1 The permissions

#	Permission	Why we need it	If denied
1	Location (while using)	GPS-stamped punch-in, check-in, order/expense location	App cannot function
2	Location (background / "always")	Route keeps recording when the phone is in the pocket	Route breaks into fragments
3	Precise location (not "approximate")	Verify the agent is <i>at</i> the shop; detect fakes	Check-in cannot be verified
4	Physical activity / motion	Our battery fix — track densely only when moving, go quiet when still	Battery drains fast
5	Run in background (foreground service)	Tracking + sync survive app being minimised	Tracking dies on screen-lock
6	Ignore battery optimisation / autostart	Stops Xiaomi/Oppo/Vivo/Realme from killing the app	App silently dies mid-day
7	Notifications	Tasks, approvals, reminders, "you forgot to punch out"	Agent misses instructions
8	Camera	Punch-in selfie, receipt photo, proof of delivery	Cannot attach proof
9	Media / photos	Store photos locally when offline; attach existing images	Photos cannot be queued
10	Phone (call)	Tap-to-call a customer or manager	No in-app calling
11	Network state	Detect when signal returns to auto-sync	(auto-granted)
12	Restart on boot	Tracking resumes if the phone is rebooted mid-shift	Tracking stops after reboot
13	Mock-location check	Block fake-GPS apps at the source	Location can be faked

Optional (only if you want the feature): Bluetooth (beacon check-in), Microphone (voice notes), Biometric (quick unlock), Contacts (avoid — privacy red flag).

2.2 How we ask (the flow that gets accepted)

Never fire a system popup cold. Explain first, then ask. A refusal is very hard to recover.

1. **Step 1 — Location:** *"Your company uses Salexia to confirm your visits and calculate your travel allowance. We only track you while you are punched in."* → ask for location.
 2. **Step 2 — Background:** *"So your route records while you drive, allow 'All the time'. You'll always see a notification when tracking is on."* → ask for background location. **Must be a separate step — the OS requires it.**
 3. **Step 3 — Activity:** *"Allow activity detection so we only use GPS when you're actually moving. This makes your battery last all day."* → ask for physical activity.
 4. **Step 4 — Battery/autostart:** brand-detected instructions + deep-link button.
 5. **Step 5 — Notifications.**
 6. **Just-in-time (not at onboarding):** Camera on first photo. Media on first attach. Phone on first call.
 7. **If denied:** a clear, non-blaming screen explaining what will not work + a one-tap **Fix it** button. Re-check at every app launch.
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PART 3 — WORKING OFFLINE (how nothing is ever lost)

3.1 What gets saved on the phone

- Attendance punches, check-in/out, visit reports, orders, payment collections, expense claims, form submissions.
- **GPS points** collected while offline.
- **Photos** (receipts, selfies, proof of delivery) — compressed, stored in the app's private folder.
- Reference data so the agent can work offline: my beat plan, my customers, the product catalogue with prices.

3.2 The outbox (how sync works)

1. Every action **saves on the phone first**, then joins a queue.
2. Each queued item carries a **unique key**, so if it is sent twice the server records it **once** — **no duplicate orders**.
3. When signal returns, the queue sends **oldest first**, retrying with growing delays on a patchy network.

4. **Photos upload last and separately**, so a large receipt image never blocks an order from syncing.
5. The **Sync Centre (M17)** always shows what is pending, what failed, and offers **Sync now**.
6. **Nothing is deleted locally until the server confirms** it received it.

3.3 Housekeeping

- Cap the local database and photo cache (e.g. 30 days).
 - Auto-purge synced items past the cap.
 - Warn the agent if the phone is nearly full.
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PART 4 — BATTERY & ACCURACY (our biggest competitive edge)

1. **Activity-aware tracking:** in a vehicle → track often. Walking → moderate. **Standing still** → **nearly stop**.
 2. **Distance filter:** record a point only after meaningful movement, not every few seconds.
 3. **Batch uploads:** send GPS points in small batches, not one network call per point.
 4. **Only while on duty:** tracking starts at punch-in and **stops at punch-out**. Never track outside working hours — better battery, better trust, better legal position.
 5. **Auto punch-out** if the agent forgets, so the app doesn't run all night.
 6. **Accuracy filter:** discard junk readings (poor accuracy, impossible jumps) before saving.
 7. **Persistent notification** while tracking — required by the OS, and it keeps us honest with the agent.
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PART 5 — PRIVACY & APP-STORE COMPLIANCE

1. **Tell the agent plainly** what is tracked, when, and why — at onboarding and in the always-visible notification.
2. **Track only during duty hours.** Technical rule *and* legal safeguard.
3. Record **consent**; publish a privacy policy inside the app.

4. **Play Store / App Store review:** background location is the most scrutinised permission there is. Both stores require a **prominent in-app disclosure before the request** plus a **demo video** of the use case. Apps get rejected here routinely — budget time for it.
5. Allow admins to **delete an ex-employee's location history** on request.
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PART 6 — HOW THE PHONE TALKS TO THE WEB APP

Phone sends up: punches, GPS points, visits, orders, payments, expenses, form answers, photos.

Phone receives down: beat plan, targets, tasks, forms, approvals, messages, push notifications.

Important: data can arrive **hours late** (agent was offline). Every record therefore carries its **true event time**, not its upload time — so a visit made at 11am shows at 11am even if it syncs at 6pm.

PART 7 — BUILD ORDER (mobile)

Phase 1 — the tracking core: Login, permission wizard, Home/punch, background location service, activity-aware battery logic, offline outbox, Sync centre, Diagnostics.

Phase 2 — attendance product: My route, attendance history, leave, notifications, profile.

Phase 3 — field sales: Beat plan, customer detail, check-in/out, visit report, take order, my orders.

Phase 4 — money & data: Collect payment, expenses with receipts, custom forms, targets, messages.

Ship Phase 1 + 2 as the "Agent Live Tracking" product. Add Phases 3 + 4 to become "Sales & Field Force".

PART 8 — SCREEN MAP

MOBILE APP

- |— Splash / auto-login
- |— Login
- |— Permission setup wizard ← 5 guided steps
- |— Home / My Day (PUNCH) ← main screen
- |— Beat plan / My route
- |— Customer detail
- |— Check-in / check-out
- |— Visit report
- |— Take order
- |— My orders
- |— Collect payment
- |— Expenses
- |— Attendance & leave
- |— Custom forms
- |— My targets
- |— Messages / notifications
- |— Sync centre ← low-network lifeline
- |— Diagnostics / self-check
- |— Profile & settings